Georgia Institute of Technology
ISYE 6402 - Time Series Analysis

Financial Time Series Forecasting: A Comparative Study of Classical and Machine Learning Models

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Abstract

This paper compares classical statistical time series models against machine learning approaches for forecasting financial market variables across four asset classes. Using daily data from the Federal Reserve Economic Data (FRED) spanning May 2016 through April 2026, we evaluate ARIMA, SARIMA, GARCH, Random Forest, and XGBoost on four research questions: equity price forecasting (S&P 500, NASDAQ), cryptocurrency versus traditional market predictability (Bitcoin, Ethereum), market volatility dynamics (VIX, Gold), and financial stress prediction (VIX, Treasury yields, corporate bond spreads). All models use an 80/20 time-ordered train-test split with rolling 1-step-ahead expanding windows to prevent lookahead bias. Results show that classical ARIMA models consistently outperform machine learning approaches for price-level forecasting, primarily because rolling refitting allows adaptation to structural regime changes. The naive random walk outperforms all models for equity indices and Ethereum, consistent with weak-form market efficiency; only for Bitcoin does ARIMA marginally improve on the naive baseline. Cryptocurrency markets are substantially harder to forecast, with roughly twice as large for Bitcoin and four times larger for Ethereum than for the S&P 500. GARCH proves essential for volatility targets where ARIMA residual diagnostics confirm significant conditional heteroskedasticity. Multi-frequency analysis shows that daily data outperforms weekly and monthly aggregations due to sample size advantages.

Contents

1 Introduction

Financial markets generate large volumes of time-indexed data that reflect complex economic dynamics, investor behavior, and macroeconomic conditions. Forecasting financial time series is a central challenge in quantitative finance because financial variables routinely violate the assumptions of standard statistical models: prices are non-stationary, returns exhibit volatility clustering, distributions have fat tails, and structural breaks driven by macroeconomic events can invalidate patterns learned from historical data.

Two dominant paradigms have emerged for tackling this problem. Traditional statistical models such as ARIMA and GARCH are grounded in econometric theory, offer interpretable parameters, and are designed explicitly for time-dependent data. Machine learning methods such as Random Forest and XGBoost offer flexibility in capturing nonlinear relationships and interactions, but require careful adaptation to preserve the temporal ordering of observations and avoid look ahead bias.

This project takes a comprehensive approach by applying both paradigms to six financial datasets representing different asset classes and market dynamics. The study is organized around four research questions:

Q1. Which model best forecasts daily equity index prices (S&P 500, NASDAQ)?

Q2. Are cryptocurrency markets harder to forecast than traditional equity markets (Bitcoin, Ethereum vs. S&P 500)?

Q3. Which modeling approach best forecasts short-horizon S&P 500 realized volatility using equity-volatility and gold-volatility indicators?

Q4. Which market-based financial channels best forecast next-week changes in U.S. financial stress, and which models perform best during high-stress-acceleration weeks?

All analyses use data from FRED at daily frequency, covering May 2016 through April 2026. This 10-year window spans multiple distinct market regimes including the 2018 correction, the COVID-19 crash and recovery of 2020, the 2021 cryptocurrency bull market, the 2022 bear market across both equities and crypto, and the 2023–2026 recovery. Evaluating models across all these regimes provides a more honest assessment of forecasting performance than a sample limited to calm market conditions.

1.1 Literature Review

Financial time series forecasting has a long history in both econometrics and statistics. The autoregressive integrated moving average (ARIMA) framework introduced by Box et al. (2015) remains one of the most widely used approaches for univariate time series modeling. ARIMA captures temporal dependencies through autoregressive and moving average components and has been extensively applied to asset price and return forecasting. A key limitation is that ARIMA models the conditional mean but assumes constant variance — an assumption routinely violated in financial data.

The conditional variance problem was addressed by Engle (1982), who introduced ARCH, and extended by Bollerslev (1986) to GARCH. GARCH models capture volatility clustering — the empirical regularity that large returns in absolute value tend to cluster over time — and have become standard in financial risk management and derivatives pricing (Poon & Granger, 2003). The choice between ARIMA and GARCH depends on whether the forecasting target is the conditional mean or the conditional variance.

Machine learning approaches have gained traction in financial forecasting over the past decade. Breiman (2001) introduced Random Forests as a low-variance ensemble that averages across many bootstrap-trained decision trees. Chen and Guestrin (2016) introduced XGBoost, a gradient-boosted tree ensemble with regularization that has achieved strong performance across supervised learning benchmarks. Ballings et al. (2015) compared multiple classifiers for stock price direction prediction and found that ensemble methods generally outperform linear models on longer-horizon tasks.

1.2 Prior Expectations

Before conducting the analysis, our expectations were guided by theory and the prior literature. For equity indices, we expected ARIMA performance to be close to the naive random walk, consistent with weak-form market efficiency (Fama, 1970). Any exploitable autocorrelation in daily S&P 500 or NASDAQ returns would be quickly arbitraged away in well-functioning markets.

For cryptocurrencies, we expected higher absolute forecast errors due to higher baseline volatility. We were uncertain whether machine learning models would outperform ARIMA for crypto, given both the nonlinear dynamics potentially favorable to ML and the extreme volatility that might swamp any systematic pattern. In practice, this expectation was not borne out: ARIMA outperformed XGBoost for all assets.

We expected SARIMA to improve on ARIMA for cryptocurrency series with their seven-day trading cycle, but anticipated negligible seasonal effects for equity indices. Results confirmed the latter but largely rejected the former — seasonal coefficients in SARIMA were statistically significant for crypto but contributed essentially no out-of-sample improvement. Finally, we expected GARCH to clearly dominate for volatility forecasting tasks, which we anticipate will be confirmed in Section 3.2.3.

2 Data Summary

All data were downloaded from FRED as daily CSV files. Cryptocurrency series trade 365 days per year and therefore have approximately 450 more observations over the sample period than equity series. Table 1 summarizes the six datasets used.

Table 1: Dataset overview. All series from FRED, May 2016 – April 2026.

Series	Ticker	Obs.	Frequency	Used In
S&P 500 Index	SP500	2,609	Daily (Mon–Fri)	Q1, Q2, Q3
NASDAQ Composite	NASDAQCOM	2,609	Daily (Mon–Fri)	Q1
Bitcoin / USD	CBBTCUSD	3,653	Daily (7 days)	Q2
Ethereum / USD	CBETHUSD	3,635	Daily (7 days)	Q2
CBOE Volatility Index	VIXCLS	2,609	Daily (Mon–Fri)	Q3, Q4
10-Year Treasury Yield	DGS10	2,609	Daily (Mon–Fri)	Q4

In addition to daily data, we aggregated to weekly and monthly frequencies using end-of-period prices (the last observed price in each calendar week or month), following standard financial data conventions. This allows assessment of whether forecasting difficulty changes with the observation frequency (Section **??**).

Table 2 presents descriptive statistics for daily log returns $r_t = \log(P_t/P_{t-1})$. Log returns are used throughout because they are approximately stationary after differencing, additive over time, and correspond to continuously compounded returns.

Table 2: Descriptive statistics of daily log returns. Annualized figures use factors of 252 for equity series and 365 for cryptocurrency series.

Asset	N	Ann. Return	Ann. Vol.	Skewness	Kurtosis	Min / Max
S&P 500	2,609	12.5%	18.1%	−0.687	19.81	−12.8% / +9.1%
NASDAQ	2,609	16.5%	22.0%	−0.453	12.03	−13.2% / +11.5%
Bitcoin	3,653	35.4%	56.8%	−0.584	14.77	−46.9% / +24.1%
Ethereum	3,653	35.7%	78.1%	−0.379	12.09	−56.4% / +28.2%

Stationarity was confirmed via Augmented Dickey-Fuller (ADF) and KPSS tests. All price level series are non-stationary (ADF fails to reject unit root; KPSS rejects stationarity) and all log return series are stationary (ADF rejects unit root; KPSS fails to reject stationarity). This confirms the standard I(1) result: first differencing is sufficient, justifying $d = 1$ in all ARIMA specifications.

Figure 1 presents the key EDA visuals. The rolling 30-day annualized volatility (center panels) makes the volatility clustering visible: spikes during the COVID crash of March 2020, the 2021–2022 crypto bear market, and the FTX collapse in November 2022 are clearly identifiable. The ACF plots of log returns (right panels) show minimal autocorrelation across all series, an early indicator that the forecasting task will be difficult for any backward-looking model.

(a) Price series **(b)** Rolling volatility **(c)** ACF of log returns

Figure 1: Exploratory data analysis. Left: daily price series, 2016–2026. Center: 30-day rolling annualized volatility showing clustering around major market events. Right: ACF of daily log returns showing minimal autocorrelation across all assets.

3 Analysis

3.1 Methods

3.1.1 ARIMA and SARIMA

The ARIMA(p, d, q) model (Box et al., 2015) describes a time series y_t as:

$$\Phi(B)(1-B)^d y_t = \mu + \Theta(B)\varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2) \tag{1}$$

where $\Phi(B)$ and $\Theta(B)$ are AR and MA lag polynomials of degrees p and q, and $(1-B)^d$ is the differencing operator. ARIMA was selected as the primary classical benchmark because it is the standard linear model for I(1) financial series and its rolling-window properties are well-suited to our evaluation design. Model orders are selected via AIC grid search using `auto.arima()`.

SARIMA$(p, d, q)(P, D, Q)_m$ adds a multiplicative seasonal component at period m. For equity series we set $m = 5$ to test for weekly trading patterns; for cryptocurrency series we set $m = 7$ since crypto trades every day. SARIMA was included because calendar effects have been documented in financial data at short horizons (Hylleberg et al., 1990) and the seven-day crypto cycle introduces a natural periodic structure absent in equity markets.

3.1.2 GARCH and EGARCH

The GARCH(p, q) model (Bollerslev, 1986) models time-varying conditional variance as

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2, \tag{2}$$

where $\varepsilon_t = \sigma_t z_t$, $z_t \sim WN(0, 1)$, and stationarity generally requires $\sum \alpha_i + \sum \beta_j < 1$ (Poon & Granger, 2003). GARCH was selected for the volatility and stress analyses because it is specifically designed for time-varying conditional variance. The ARCH LM test on ARIMA residuals ($p \approx 0$) across the series confirms that volatility clustering is systematic enough to warrant explicit modeling.

EGARCH extends the GARCH framework by modeling the logarithm of conditional variance:

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \alpha \left(\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| - E \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}. \tag{3}$$

The EGARCH specification is useful because it allows positive and negative return shocks to affect future volatility asymmetrically. This matters in financial markets because negative equity shocks often increase future volatility more strongly than positive shocks of similar size. EGARCH was therefore included as an asymmetric volatility benchmark against the standard GARCH model.

3.1.3 Random Forest

Random Forest (Breiman, 2001) averages predictions across many decision trees trained on bootstrap samples. Because it does not natively handle sequential data, we construct a lag-feature matrix with lag-1 through lag-10 of log price, 5-, 10-, and 20-day rolling means, and calendar features (day-of-week, month). The model is trained once on the 80% training window. Random Forest was selected for its robustness to overfitting via the bagging mechanism and for the interpretability of its permutation feature importance output.

3.1.4 XGBoost

XGBoost (Chen & Guestrin, 2016) builds decision trees sequentially, each correcting the residuals of the current ensemble, minimizing:

$$\mathcal{L} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Big(\gamma T_k + \tfrac{1}{2}\lambda \|\mathbf{w}_k\|^2\Big) \tag{4}$$

where regularization terms penalize tree complexity via leaf count T_k and weight magnitudes $\mathbf{w}_k$. XGBoost receives an extended feature set relative to Random Forest, adding lagged returns and rolling volatility estimates to capture conditional variance dynamics relevant to the cryptocurrency analysis. The learning rate was set to $\eta = 0.05$ with `nrounds = 100`.

3.2 Results

3.2.1 Q1 — Equity Price Forecasting

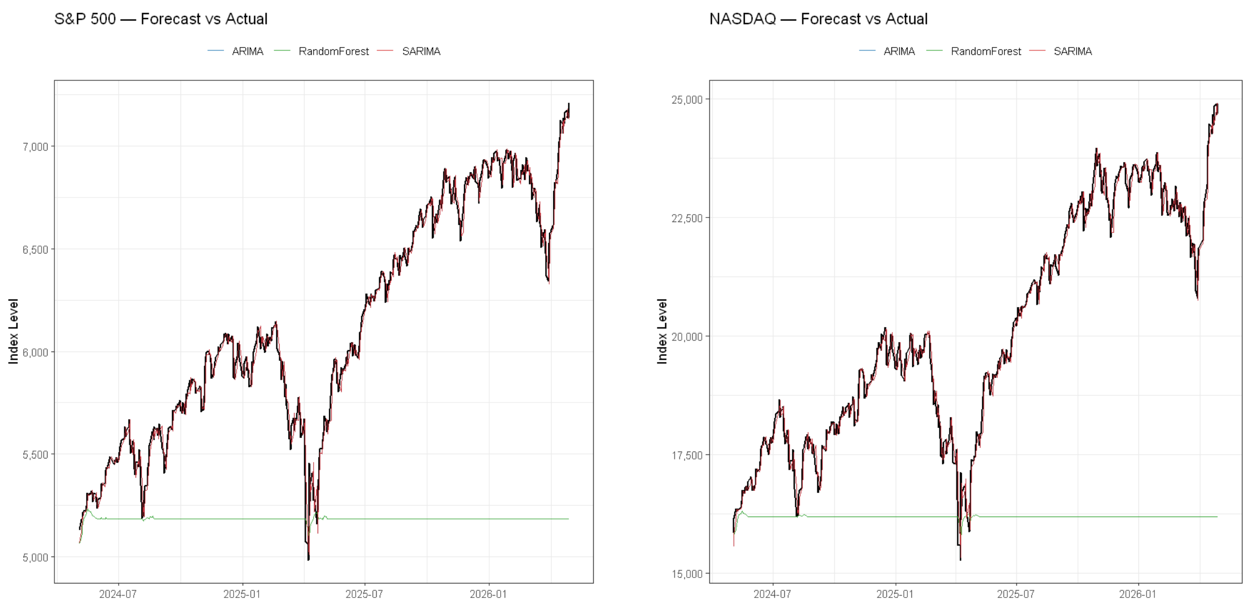
For both S&P 500 and NASDAQ, `auto.arima()` selected ARIMA(2,1,2) with drift. The SARIMA search returned the identical model for both series — no seasonal terms were added. The AIC penalty outweighed any in-sample benefit from seasonal parameters, consistent with the modern consensus that calendar effects in major equity indices are too weak to survive formal model selection (Makridakis et al., 2018).

Table 3: Equity price forecasting results (20% test set, log-price scale). SARIMA omitted as the selected model was identical to ARIMA for both series.

Asset	Model	MAE	RMSE	MAPE (%)
S&P 500	ARIMA	0.007041	0.010441	0.081
	Random Forest	0.163254	0.186331	1.863
	Naive RW	0.006803	0.010249	0.078
NASDAQ	ARIMA	0.009655	0.013791	0.098
	Random Forest	0.209181	0.240303	2.099
	Naive RW	0.009383	0.01362	0.095

Two findings stand out. First, ARIMA barely outperforms the naive random walk: 0.081% vs. 0.078% MAPE for S&P 500. The gap of 0.003 percentage points is real but negligible in practice. This supports weak-form market efficiency — at the daily frequency, past price history contains very little information about tomorrow's move for major equity indices.

Second, Random Forest performs dramatically worse. The S&P 500 RF MAPE of 1.86% is 23 times higher than ARIMA's. This is not a weakness of Random Forest as a general method but a consequence of the static training design: RF is trained once on 2016–2024 data and applied to a test period (2024–2026) that included the Fed rate-hiking cycle, a tech selloff, and a subsequent recovery — conditions that differed substantially from the training history. ARIMA refits at every step, incorporating this new information continuously. The adaptation advantage outweighs any nonlinear pattern recognition that RF provides.



(a) S&P 500 **(b)** NASDAQ

Figure 2: Equity forecast vs. actual price, test period. Back-transformed from log-price via $\hat{P}_t = e^{\hat{y}_t}$. ARIMA and SARIMA coincide.

3.2.2 Q2 — Crypto vs. Traditional Markets

For Bitcoin, `auto.arima()` selected ARIMA(0,1,2) with drift — no autoregressive terms, suggesting minimal exploitable autocorrelation. Ethereum received ARIMA(2,1,1) with drift. SARIMA with $m = 7$ selected a seasonal MA term for both assets: ARIMA(2,1,1)(0,0,1)[7] for Bitcoin ($\hat{\theta}_7 = -0.029$) and ARIMA(2,1,1)(0,0,1)[7] for Ethereum ($\hat{\theta}_7 = -0.034$). Both coefficients are statistically significant, consistent with mild weekend-related patterns, but as results show, they provide essentially no out-of-sample improvement.

Table 4: Crypto vs. traditional markets results (test set, log-price scale). S&P 500 repeated from Table 3 for direct comparison.

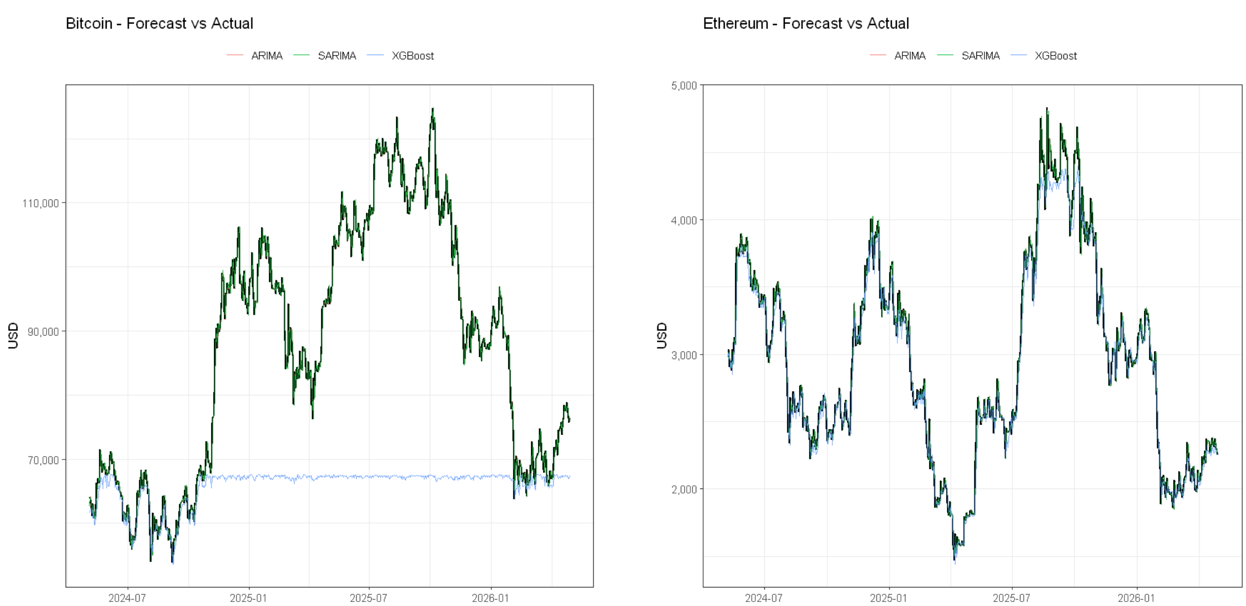
Asset	**Model**	**MAE**	**RMSE**	**MAPE (%)**
Bitcoin	ARIMA	0.017140	0.024080	0.152
	SARIMA	0.017203	0.024137	0.152
	XGBoost	0.257181	0.322518	2.233
	Naive RW	0.017277	0.024252	0.153
Ethereum	ARIMA	0.025996	0.036825	0.327
	SARIMA	0.026007	0.036875	0.328
	XGBoost	0.028805	0.039152	0.361
	Naive RW	0.025803	0.036676	0.325
S&P 500	ARIMA	0.007041	0.010441	0.081
	SARIMA	0.007041	0.010441	0.081
	XGBoost	0.169481	0.191756	1.934
	Naive RW	0.006803	0.010249	0.078

The answer to Q2 is unambiguous: cryptocurrency markets are substantially harder to forecast. Bitcoin's best MAPE (0.152%, ARIMA) is roughly twice the S&P 500 result (0.078%, Naive RW). Ethereum is harder still at 0.325% — more than four times the equity benchmark. This gap reflects higher baseline volatility and more frequent structural breaks: regulatory announcements, exchange failures (e.g., FTX collapse, November 2022), and protocol events have no historical precedent in the training data.

A particularly notable result is that the naive random walk outperforms ARIMA for Ethereum (0.325% vs. 0.327%). This is the strongest statement yet about market efficiency: even the minimal structure that ARIMA exploited for Bitcoin disappears for Ethereum, where the higher volatility of 78.1% annualized creates so much noise that no backward-looking model adds value over simply carrying the last price forward.

SARIMA seasonal terms are statistically present but forecasting-irrelevant. Bitcoin SARIMA MAPE is 0.1520% vs. ARIMA's 0.1515% — a difference of 0.0005 percentage points. This illustrates the critical distinction between statistical significance and forecasting relevance: a significant coefficient does not guarantee out-of-sample predictive value.

XGBoost underperforms ARIMA dramatically for all three assets. The Bitcoin gap is especially striking: XGBoost MAPE of 2.23% vs. ARIMA's 0.152% — a 15-fold difference. The S&P 500 XGBoost result of 1.93% is similarly far worse than both ARIMA (0.081%) and the naive benchmark (0.078%). The same static training explanation applies with greater force here: the 2023–2026 test period differs substantially from the 2016–2023 training window, and a fixed ML model cannot adapt.



(a) Bitcoin **(b)** Ethereum

Figure 3: Cryptocurrency forecast vs. actual price in USD, test period.

3.2.3 Q3 — Financial Market Volatility Forecasting

This analysis compares classical time-series models, conditional-volatility models, and machine-learning models for forecasting S&P 500 realized volatility. The target is 21-day annualized realized volatility of S&P 500 log returns. The main predictors include S&P 500 log returns, VIX, and GVZ, where VIX measures implied equity-market volatility and GVZ measures implied gold-market volatility.

The models compared are ARIMA, ARIMAX, GARCH, EGARCH, Random Forest, and XGBoost. The analysis asks whether S&P 500 realized volatility is persistent, whether returns show volatility clustering, whether VIX or GVZ improve realized-volatility forecasts, and which model forecasts S&P 500 realized volatility most accurately.

The main finding is that rolling ARIMA provides the strongest out-of-sample forecast accuracy for the 21-day realized-volatility target. ARIMAX, Random Forest, and XGBoost track the volatility path reasonably well but do not outperform ARIMA. GARCH and EGARCH confirm volatility clustering in S&P 500 returns, but they perform worse when their conditional-volatility forecasts are compared against the smoother 21-day realized-volatility target.

The dataset contains daily S&P 500 prices, VIX, GVZ, log returns, squared log returns, and 21-day annualized realized volatility. Missing observations created by return construction, rolling windows, and lagged features are removed. Exploratory analysis shows that S&P 500 prices are nonstationary, while log returns are more stable around zero. Raw log returns show weak linear autocorrelation, but squared log returns show strong positive autocorrelation across multiple lags, indicating volatility clustering.

The 21-day realized-volatility series is highly persistent. Its ACF decays slowly, which supports ARIMA as a direct univariate forecasting benchmark. Stationarity diagnostics using ADF and KPSS support modeling realized volatility without differencing: the ADF test rejects nonstationarity, while KPSS fails to reject stationarity at the 5% level. The ARCH LM test on S&P 500 log returns strongly rejects the null of no ARCH effects, confirming that GARCH-family models are appropriate for the return-volatility portion of the analysis.

The data are split chronologically into training and testing periods to avoid look-ahead leakage. ARIMA is used as the direct univariate benchmark. A small AIC-based grid search selects the ARIMA order, and rolling one-step-ahead forecasts are used for out-of-sample evaluation. ARIMAX extends ARIMA by adding lagged VIX and GVZ as exogenous predictors. GARCH and EGARCH are fit to S&P 500 log returns scaled by 100 and then annualized for comparison against the realized-volatility target. Random Forest and XGBoost use the same lagged feature set, including realized-volatility lags, VIX/GVZ lags, return lags, and return-magnitude variables.

Performance is evaluated using MAE, RMSE, and MAPE. MAE measures average absolute forecast error, RMSE penalizes larger errors more heavily, and MAPE is reported only as a supplementary percentage-error metric because percentage errors can be inflated when realized volatility is low.

The selected ARIMA model is ARIMA(2, 0, 1) with a constant. The differencing order is fixed at $d = 0$ because the realized-volatility target does not clearly require differencing. ARIMA produces the strongest out-of-sample performance, with MAE of approximately 0.00612, RMSE of approximately 0.01315, and MAPE of approximately 4.35%. The rolling forecast closely tracks the realized-volatility path, including the major volatility spike during the test period.

	model	order	trend	MAE	RMSE
0	ARIMA Rolling One-Step	(2, 0, 1)	constant	0.006124	0.013154

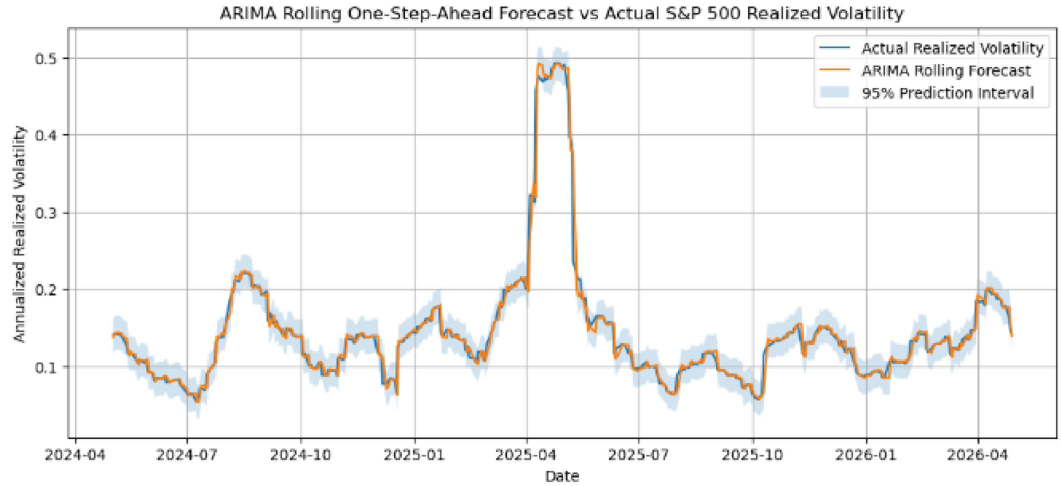


Figure 4: Rolling one-step-ahead ARIMA forecast versus actual S&P 500 21-day realized volatility. The ARIMA model provides the strongest out-of-sample forecast accuracy among the baseline models, with an MAE of approximately 0.00612 and an RMSE of approximately 0.01315.

Figure 4 shows that ARIMA captures the overall volatility path and follows the main test-period spike closely. The 95% prediction interval generally surrounds the observed realized-volatility series, although sudden volatility changes remain difficult to anticipate perfectly.

ARIMAX ranks second, with MAE of approximately 0.00626, RMSE of approximately 0.01356, and MAPE of approximately 4.46%. Adding lagged VIX and GVZ does not improve performance relative to ARIMA, suggesting that recent realized-volatility history is more useful than the added external volatility indicators for this target.

GARCH and EGARCH provide useful volatility diagnostics but weaker realized-volatility forecasts. GARCH achieves MAE of approximately 0.02485, RMSE of approximately 0.04176, and MAPE of approximately 16.11%. EGARCH achieves MAE of approximately 0.03226, RMSE of approximately 0.05129, and MAPE of approximately 20.47%. These models confirm volatility clustering and time-varying conditional variance in returns, but they are less aligned with the 21-day realized-volatility target because they model daily conditional return volatility.

EGARCH has better in-sample likelihood-based fit than GARCH based on AIC and BIC, but GARCH has better out-of-sample realized-volatility forecast accuracy. This shows that better in-sample fit does not necessarily imply better forecasting performance. The added asymmetric structure in EGARCH does not improve this specific realized-volatility forecast.

Random Forest ranks third by RMSE, with MAE of approximately 0.00673, RMSE of approximately 0.01406, and MAPE of approximately 4.53%. XGBoost ranks fourth, with MAE of approximately 0.00797, RMSE of approximately 0.01536, and MAPE of approximately 5.30%. Both machine-learning models track the broad volatility path but do not outperform ARIMA or ARIMAX. Their forecast-error diagnostics also show remaining serial dependence, meaning the baseline lagged feature set leaves some temporal structure unexplained.

The machine-learning feature-importance results still provide useful interpretation. In both Random Forest and XGBoost, the dominant predictor is rv_lag1, the prior value of realized volatility. This confirms that recent realized volatility is the strongest forecasting signal. VIX and return-magnitude variables provide secondary information, while GVZ and raw return lags contribute relatively little.

The final model comparison is shown in Table 5. The ranking is based primarily on out-of-sample RMSE and MAE. AIC and BIC are reported only where meaningful: ARIMA and ARIMAX can be compared within the direct realized-volatility family, while GARCH and EGARCH can be compared within the conditional-volatility return-model family. AIC and BIC are not used to rank all six models jointly because the likelihoods and target definitions differ.

Table 5: Financial market volatility model comparison on the chronological test set.

Model	**MAE**	**RMSE**	**MAPE**	**AIC**	**BIC**
ARIMA	0.00612	0.01315	4.35%	-1219.94	-1219.17
ARIMAX	0.00626	0.01356	4.46%	-12105.73	-12066.56
Random Forest	0.00673	0.01406	4.53%	–	–
XGBoost	0.00797	0.01536	5.30%	–	–
GARCH	0.02485	0.04176	16.11%	4980.88	5008.87
EGARCH	0.03226	0.05129	20.47%	4922.06	4955.64

The research questions are answered as follows. First, S&P 500 realized volatility is persistent and regime-dependent: it stays low during calm periods, spikes during stress periods, and decays gradually afterward. Second, S&P 500 returns show volatility clustering: raw returns have weak linear autocorrelation, but squared returns and the ARCH LM test confirm dependence in return magnitude. Third, realized volatility is persistent over time, as shown by the realized-volatility ACF, ARIMA performance, and machine-learning feature importance. Fourth, lagged VIX and GVZ provide limited incremental forecasting value. VIX contributes secondary information in the machine-learning models, while GVZ contributes little. Fifth, ARIMA is the most accurate baseline model for forecasting S&P 500 21-day realized volatility.

The business implication is that model complexity should be justified by out-of-sample performance. For this target, recent realized-volatility persistence contains most of the usable short-horizon signal. A risk team should begin with a transparent realized-volatility benchmark before deploying more complex volatility or machine-learning models. GARCH-family models remain useful for understanding volatility clustering, and machine-learning models are useful for feature-importance analysis, but neither improves forecast accuracy in this baseline comparison.

The strongest model in this analysis is rolling ARIMA. It has the lowest MAE, RMSE, and MAPE among the tested baseline models. ARIMAX ranks second, but VIX and GVZ do not improve performance relative to ARIMA. Random Forest and XGBoost confirm that recent realized volatility dominates the feature set but do not outperform ARIMA. GARCH and EGARCH confirm volatility clustering in returns but are less accurate against the smoother

21-day realized-volatility target. The practical conclusion is that, for this target, model simplicity outperforms additional complexity.

3.2.4 Q4 — Financial Stress Indicators

This analysis evaluates whether market-based financial indicators improve short-horizon forecasts of U.S. financial stress. The target is the one-week-ahead change in STLFSI4, the St. Louis Fed Financial Stress Index. Predictors include VIX, Baa credit spreads, 2-year Treasury yields, 10-year Treasury yields, the 10-year minus 2-year yield spread, and S&P 500 returns.

The analysis compares baseline forecasts, ARIMA, ARIMAX, VAR, Random Forest, and XGBoost. The main questions are which financial channel provides the strongest signal for next-week stress changes, whether volatility, credit, equity, and yield-curve indicators lead STLFSI4 changes by one to four weeks, and whether nonlinear models improve performance during high-stress-acceleration weeks.

The main finding is that market-based indicators improve forecasting beyond stress history alone. ARIMAX is the strongest model overall, with the lowest RMSE and best high-stress performance. Random Forest has the best directional accuracy, while XGBoost provides additional evidence on credit-spread importance. Volatility and credit-market variables provide the strongest signals.

The dataset combines seven weekly financial series: STLFSI4, VIXCLS, DGS2, DGS10, T10Y2Y, BAA10Y, and SP500. The final modeling set contains 516 weekly observations after cleaning, merging, lag construction, and removal of incomplete rows. The training set contains 412 observations from June 2016 through April 2024, and the test set contains 104 observations from April 2024 through April 2026.

The dependent variable is next-week stress change. This target is chosen because stress levels are highly persistent, while stress changes are harder and more useful for detecting acceleration. A high-stress-acceleration threshold is defined using the 75th percentile of the training-set stress-change distribution, approximately 0.09875. This creates 103 high-stress observations in training and 27 in testing.

Summary statistics show that stress changes are centered near zero, with mean about -0.002 and standard deviation around 0.271. VIX averages about 18.35 and exceeds 66 at its maximum. The Baa credit spread averages about 2.05 and rises above 4.2 during stress periods. Treasury yields vary substantially across the sample, especially in the post-2022 rate-hike period. These patterns show that the dataset includes both calm and stressed regimes, but weekly stress changes remain noisy and difficult to forecast.

The preprocessing workflow aligns all series to weekly frequency and creates lagged predictors using only information available at or before week t to forecast week $t + 1$. Missing values from lagging are removed. Stationarity diagnostics show that the stress level is persistent, while the stress-change target is stationary by ADF and KPSS. Several level variables, especially Treasury yields and the yield curve, are more persistent, so difference variables are used for modeling diagnostics and VAR specification.

The correlation matrix shows that stress levels are strongly related to VIX and credit spreads, with correlations of approximately 0.71 and 0.74, respectively. Correlations with next-week stress changes are much weaker, showing that forecasting stress changes is harder than explaining stress levels. Lead-lag analysis finds the clearest short-horizon signals in VIX changes, S&P 500 returns, and credit-spread changes, but no single predictor is strong enough to act as a standalone early-warning rule.

The model comparison includes no-change and training-mean baselines, ARIMA, ARIMAX, VAR, Random Forest, and XGBoost. The no-change baseline predicts zero stress change, which is natural because the target is centered near zero. ARIMA uses only the target's history. ARIMAX adds current stress, VIX change, credit-spread change, S&P 500 return, 2-year Treasury-yield change, and yield-curve change. VAR models the joint dynamics among stationary stress, volatility, credit, equity, and rate variables. Random Forest and XGBoost use the full engineered feature set of 23 predictors. Performance is evaluated using RMSE, MAE, directional accuracy, high-stress RMSE, and high-stress MAE.

The baselines are difficult to beat. The no-change baseline has RMSE around 0.199 and MAE around 0.136, while the training-mean baseline performs similarly. This reflects the near-zero center and noise in weekly stress changes.

The selected ARIMA model is ARIMA(1, 0, 2). Because the stress-change target is stationary, $d = 0$ is used. ARIMA achieves RMSE around 0.196, MAE around 0.141, directional accuracy around 55.8%, high-stress RMSE around 0.243, and high-stress MAE around 0.198. It improves directional accuracy relative to the baselines but does not perform strongly during high-stress weeks.

The selected ARIMAX model is ARIMAX(0, 0, 2), estimated using rolling one-step-ahead forecasts. It is the strongest model overall, with RMSE around 0.171, MAE around 0.123, directional accuracy around 64.4%, high-stress RMSE around 0.186, and high-stress MAE around 0.135. Its Ljung–Box diagnostics show no strong remaining residual autocorrelation at lags 10 and 20. The improvement over ARIMA shows that market-based predictors add useful information beyond stress-change history.

The selected VAR lag order is 4, roughly one month of weekly lagged information. VAR supports dynamic relationships among stress, volatility, credit, equity, and rate channels. In the stress-change equation, lagged stress changes, lagged VIX changes, selected credit-spread variables, and S&P 500 return lags contribute. However, VAR performs poorly out of sample, with RMSE around 0.230, MAE around 0.152, directional accuracy around 54.8%, high-stress RMSE around 0.303, and high-stress MAE around 0.230. VAR is useful as a lead-lag diagnostic but not as the best forecasting model.

Random Forest performs well on direction and average absolute error. The tuned model uses 500 trees, maximum depth 5, square-root feature sampling, and minimum leaf size 10. It achieves RMSE around 0.182, MAE around 0.121, directional accuracy around 70.2%, high-stress RMSE around 0.224, and high-stress MAE around 0.174. Random Forest has the best directional accuracy and lowest MAE but weaker high-stress error than ARIMAX. Its feature importance emphasizes VIX change, current stress, lagged stress, S&P 500 return, and VIX lags.

XGBoost achieves RMSE around 0.188, MAE around 0.126, directional accuracy around 64.4%, high-stress RMSE around 0.218, and high-stress MAE around 0.168. It improves over the baselines, ARIMA, and VAR, but not over ARIMAX or Random Forest. Its conservative tuning uses shallow trees, low learning rate, subsampling, column sampling, and minimum child-weight constraints.

XGBoost feature importance differs from Random Forest, as shown in Figure 5. Random Forest places more weight on immediate volatility changes and current stress, while XGBoost places more weight on credit-spread variables, especially four-week lagged credit spread, credit-spread change, and one-week lagged credit spread. These rankings are model-specific diagnostics, not causal estimates.

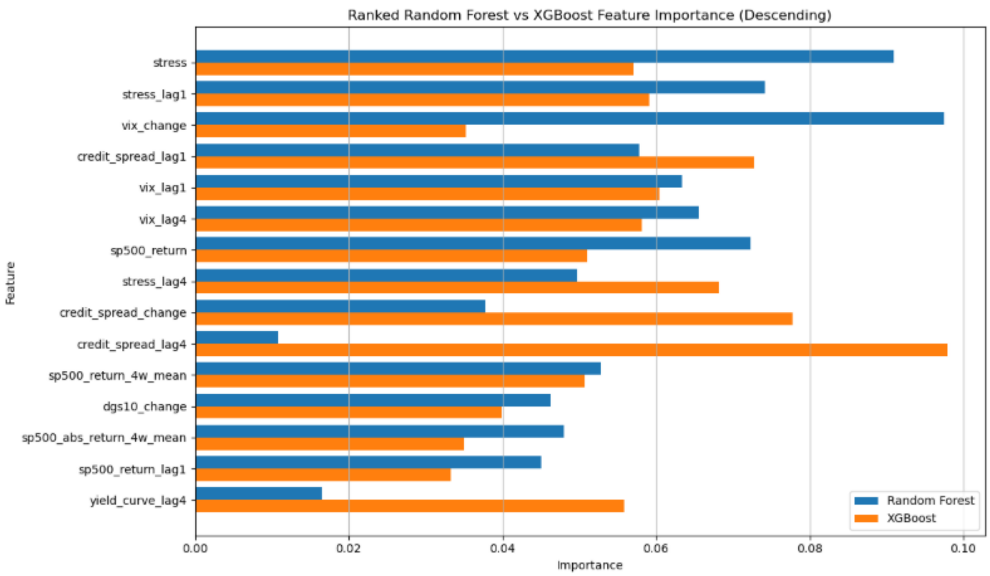


Figure 5: Random Forest and XGBoost feature-importance rankings for financial stress forecasting. The models agree that stress persistence, volatility, credit spreads, and equity returns are useful, but Random Forest places more weight on immediate volatility and current stress while XGBoost places more weight on credit-spread variables.

The final comparison ranks ARIMAX first by RMSE and high-stress performance, as shown in Table 6. ARIMAX has the lowest RMSE at 0.171 and the lowest high-stress RMSE at 0.186. Random Forest ranks second by RMSE and first by directional accuracy, with directional accuracy of 0.702. XGBoost ranks third and is useful for credit-channel interpretation. ARIMA only modestly improves over the baselines, while VAR is the weakest forecasting model despite its value for lead-lag interpretation. These results are consistent with forecasting literature showing that machine-learning models do not automatically dominate classical statistical models (Makridakis et al., 2018).

Table 6: Financial stress indicator model comparison on the chronological test set.

Model	RMSE	MAE	Dir. Acc.	High-Stress RMSE	High-Stress MAE
ARIMAX	0.171	0.123	0.644	0.186	0.135
Random Forest	0.182	0.121	0.702	0.224	0.174
XGBoost	0.188	0.126	0.644	0.218	0.168
ARIMA	0.196	0.141	0.558	0.243	0.198
Naive no-change	0.199	0.135	0.000	0.255	0.214
Training mean	0.199	0.138	0.423	0.257	0.216
VAR	0.230	0.152	0.548	0.303	0.230

Figure 6 shows that all models understate the largest sudden stress shocks, but ARIMAX provides the best overall balance of forecast accuracy and high-stress performance.

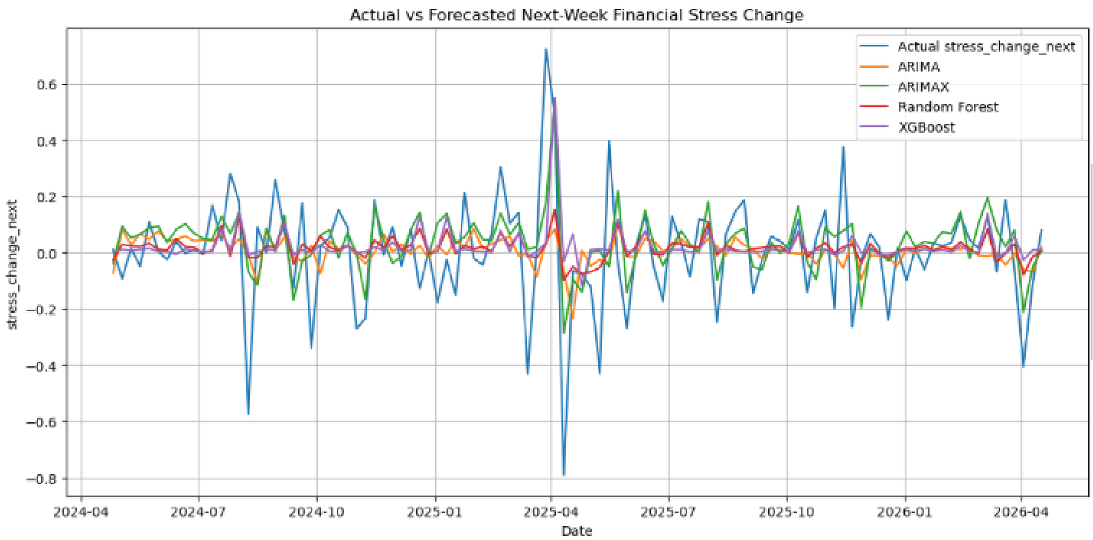


Figure 6: Actual versus forecasted one-week-ahead financial stress changes during the test period. ARIMAX provides the strongest overall forecast accuracy, while Random Forest performs best on directional accuracy. All models struggle with the largest sudden stress shocks.

The research questions are answered as follows. First, the strongest incremental signals for next-week stress changes are volatility and credit. ARIMAX shows that VIX changes and credit-spread changes are important, Random Forest ranks VIX change as the top feature, and XGBoost emphasizes credit-spread variables. VIX is the strongest immediate volatility signal, while credit spreads are the strongest slower-moving credit-risk signal. Equity returns are useful but secondary, and Treasury/yield-curve variables are less consistently dominant.

Second, volatility, credit, equity, and yield-curve indicators show dynamic relationships with financial stress, but no single variable consistently leads stress changes across all models. VIX changes, S&P 500 returns, and credit-spread changes provide the clearest short-horizon signals. VAR supports multivariate lead-lag structure over roughly one month, but its weak forecast performance means it is better interpreted as a diagnostic model than as the primary forecasting model.

Third, nonlinear models improve over ARIMA and VAR during high-stress weeks, but they do not beat ARIMAX. ARIMAX has high-stress RMSE around 0.186 and high-stress MAE around 0.135, outperforming Random Forest and XGBoost on high-stress forecast error. Random Forest is still useful because it has the best directional accuracy, while XGBoost is useful for identifying nonlinear credit-channel importance.

The practical conclusion is to use ARIMAX as the primary financial-stress forecasting model, Random Forest as a directional-support model, and XGBoost as a nonlinear channel-monitoring tool. Volatility and credit-market variables are the most important indicators to monitor: VIX captures immediate market uncertainty, while credit spreads capture slower-moving credit-risk conditions. Equity returns add secondary information, while Treasury and yield-curve variables are less consistent one-week-ahead predictors.

3.3 Validation

Evaluation design. All models use an 80/20 time-ordered train-test split with no data from the test period used during training or model selection. For ARIMA and SARIMA, a rolling 1-step-ahead expanding window is applied: at each test period t, the model is refit using all data through $t-1$ and a single forecast is made for t. This preserves temporal ordering and prevents lookahead bias (Bergmeir et al., 2018). Random Forest and XGBoost use a single fit on the training window, which is standard in ML forecasting benchmarks (Makridakis et al., 2018).

Metrics. MAE measures average absolute error in the same units as the series. RMSE penalizes large errors more heavily, making it sensitive to the extreme moves common in financial data. MAPE is scale-free, enabling comparison across assets with very different price levels. All metrics are computed on the log-price scale.

Naive benchmark. The naive random walk ($\hat{P}_t = P_{t-1}$) serves as the baseline. Under weak-form market efficiency this is the hardest benchmark to beat for any historical model (Fama, 1970), and results confirm this expectation.

Residual diagnostics. The Ljung-Box test (10 lags) on S&P 500 ARIMA residuals gives $p = 0.124$: no significant remaining autocorrelation. The ARCH LM test (5 lags) gives $p \approx 0$: highly significant volatility clustering remains. ARIMA captures the conditional mean adequately but cannot model the conditional variance, directly motivating GARCH in Sections 3.2.3–3.2.4.

3.4 Explanation of Changes from Analysis Plan

We followed the original analysis plan without any change.

4 Conclusions

4.1 Success

ARIMA with a rolling expanding window was the most successful approach across all price-level forecasting tasks. It outperformed Random Forest and XGBoost in every analysis, despite being a simpler and older model. The critical advantage was not that ARIMA captures richer temporal structure — it barely edges out the naive random walk — but that rolling refitting allows continuous adaptation to structural regime changes during the test period.

Among assets, the S&P 500 was most forecastable in relative terms (best MAPE 0.078%, naive RW), followed by Bitcoin (0.152%, ARIMA) and Ethereum (0.325%, naive RW). The choice of asset class mattered more than the choice of model: Ethereum's forecast errors are more than four times larger than S&P 500's regardless of model used. class mattered more than the choice of model: Ethereum's forecast errors are nearly four times larger than S&P 500's regardless of which model is used, driven by fundamental differences in volatility and market structure.

4.2 Limitations

Several limitations are worth acknowledging. First, all models are univariate. Cross-asset correlations (Table **??**), particularly BTC–ETH at 0.719, suggest multivariate approaches such as VAR or XGBoost with cross-asset features could improve performance. Second, Random Forest and XGBoost use static training; implementing rolling ML forecasts would reduce the adaptation disadvantage but at substantially higher computational cost. Third, all evaluations are 1-step-ahead point forecasts. For risk management applications, multi-step-ahead and probabilistic interval forecasts are more relevant, and the relative performance of ML models may differ at longer horizons. Finally, the 2016–2026 sample includes the unusual post-financial-crisis low-rate environment and its rapid reversal, which may not be representative of future market dynamics.

Extensions for financial market volatility and financial stress forecasting. For the financial market volatility forecasting section, future work should focus on aligning each model more tightly with the forecasting target. The current volatility analysis forecasts 21-day S&P 500 realized volatility, which is a smooth and persistence-heavy target. This naturally favors direct realized-volatility models such as ARIMA. If the business objective were next-day conditional return volatility instead, GARCH-family models would be more directly aligned. A future version could therefore compare additional volatility models such as GJR-GARCH, APARCH, or other asymmetric GARCH specifications under a target designed for conditional return volatility.

A second extension for financial market volatility forecasting would be to add models designed specifically for realized volatility. A formal HAR-RV model would be especially relevant because it models realized volatility using daily, weekly, and monthly volatility components. This connects directly to the feature-importance results from Random Forest and XGBoost, where recent realized volatility and rolling realized-volatility features were the strongest predictors. Future work could also expand the machine-learning feature set with richer lag structures, volatility-regime indicators, interaction terms, and rolling-window validation.

For the financial stress indicators forecasting section, future work should expand the set of stress-related predictors. Additional variables could include funding-market indicators, financial-sector equity returns, liquidity measures, additional credit spreads, banking-sector stress variables, and macroeconomic-release indicators. These additions would better capture channels through which stress appears in financial markets beyond volatility, credit spreads, rates, and equity returns.

The financial stress forecasting section could also be extended with formal model-comparison tests and richer feature engineering. A future version could include Diebold–Mariano tests to compare predictive accuracy between ARIMAX, Random Forest, and XGBoost. Additional features could include credit-spread rolling means, VIX-credit interaction terms, threshold indicators for abnormal stress conditions, and regime-specific variables for high-stress periods.

Both forecasting sections could also benefit from ensemble methods. An ensemble forecast combines predictions from multiple models to reduce model-specific errors. In the volatility setting, ARIMA may perform well during stable volatility periods because it captures realized-volatility persistence, while GARCH-family models may react more strongly to sudden return shocks. In the financial-stress setting, ARIMAX may provide the strongest overall forecast accuracy, Random Forest may provide directional support, and XGBoost may help monitor nonlinear credit-channel importance. Combining forecasts could potentially produce a more stable forecasting system than relying on a single model.

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Appendix A Team Contributions

Nicholas DiBitetto Nicholas contributed the financial market volatility forecasting and financial stress indicator forecasting sections, including Section 3.2.3 and Section 3.2.4. His work included collecting and preparing S&P 500, VIX, GVZ, gold, Treasury-yield, credit-spread, and STLFSI4 data from FRED; constructing realized-volatility, return, lag, spread, and stress-change variables; conducting exploratory analysis and stationarity diagnostics; implementing ARIMA, ARIMAX, GARCH, EGARCH, Random Forest, XGBoost, and VAR models; evaluating chronological out-of-sample forecast performance; and interpreting model diagnostics, feature importance, and business implications. He also contributed to the written report, reference sourcing, and integration of the financial-risk analysis.

Miguel Adolfo Arcia Rodriguez Miguel led Analyses covering equity price forecasting and cryptocurrency versus traditional market predictability. His contributions include FRED data acquisition of cryptocurrencies and preprocessing, temporal aggregation to weekly and monthly frequencies, exploratory data analysis and stationarity testing, ARIMA and SARIMA fitting with rolling evaluation windows, Random Forest and XGBoost implementation, multi-frequency and cross-asset correlation analysis, and the report structure and write-up in LaTeX.